



Ext 1

## NORTH SYDNEY BOYS HIGH SCHOOL

2009  
TRIAL HIGHER SCHOOL CERTIFICATE  
EXAMINATION

# Mathematics

## Extension 1

### General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write on one side of the paper (with lines) in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a new page.

- Attempt all questions

#### Class Teacher:

(Please tick or highlight)

- ☐ Mr Barrett
- ☐ Mr Ireland
- ☐ Mr Lowe
- ☐ Mr Fletcher
- ☐ Mr Trenwith
- ☐ Mr Weiss

Student Number:

(To be used by the exam markers only.)

| Question No | 1               | 2               | 3               | 4               | 5               | 6               | 7               | Total           | Total            |
|-------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|
| Mark        | $\overline{12}$ | $\overline{12}$ | $\overline{12}$ | $\overline{12}$ | $\overline{12}$ | $\overline{12}$ | $\overline{12}$ | $\overline{84}$ | $\overline{100}$ |

**Question 1 (12 marks)****Marks**

- (a) Find  $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$  1
- (b) Differentiate  $\log_e(\sin^3 x)$ , writing the answer in simplified form. 2
- (c) Find the range of the function  $f(x) = x^2 - 6x + 10$ . 2
- (d) Find the acute angle between the lines  $-x + 2y + 4 = 0$  and  $3x + y + 1 = 0$ . Give your answer to the nearest minute. 2
- (e) Evaluate  $\int_1^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}}$  2
- (f) i) Show that  $x-2$  is a factor of  $x^3 - 4x^2 + 7x - 6$  1  
ii) Show why  $x^3 - 4x^2 + 7x - 6 = 0$  has only 1 real root. 2

**Question 2 (12 marks)**

- (a) Find the exact value of  $\cos \left[ \sin^{-1} \left( \frac{1}{3} \right) \right]$  2
- (b) Solve the equation  $3x^3 - 17x^2 - 8x + 12 = 0$  given that the product of two of the roots is 4. 3
- (c)
- (i) Express  $\sqrt{12} \sin x + 2 \cos x$  in the form  $R \cos (x - \alpha)$  where  $R > 0$  and  $0 \leq x \leq \frac{\pi}{2}$  2
- (ii) Hence solve  $\sqrt{12} \sin x + 2 \cos x = -3$  for  $0 \leq x \leq 2\pi$  2
- (d) Solve the inequality  $\frac{x^2 - 9}{x} \leq 8$  3

**Question 3 (12 marks)**

- (a) (i) Show why the equation  $\ln(x+1) + x - 1 = 0$  must have a root between  $x=0$  and  $x=1$ . 1
- (ii) Given that the solution of  $\ln(x+1) + x - 1 = 0$  is approximately equal to  $\frac{1}{2}$ , use one application of Newton's method to get a better approximation. Write your answer correct to two decimal places. 2
- (b) The acceleration of a particle  $P$  is given by  $a = 32x(x^2 + 1)$  where  $x$  cm is the displacement at time  $t$  sec. Initially  $P$  starts from the origin with velocity 4 cm/s.
- (i) Use the fact that  $\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$  to show that  $v = \pm 4(x^2 + 1)$  2
- (ii) Justify why the positive solution is the correct one. 1
- (iii) Find  $x$  in terms of  $t$ . 2
- (c) Prove using mathematical induction that, for all positive integers  $n \geq 1$ ,

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4n-3) \times (4n+1)} = \frac{n}{4n+1} \quad 4$$

**Question 4 (12 marks)**

- (a) Find  $\int \cos^2 3x \, dx$  2
- (b) (i) Show that the maximum value of  $2x(1-x)$  is  $\frac{1}{2}$  1
- (ii) Find the range of the function  $f(x) = \sin^{-1}\{2x(1-x)\}$ , with domain  $0 \leq x \leq 1$ . 2
- (c) (i) Show that  $T = 22 + Ae^{kt}$  is a solution to the equation  $\frac{dT}{dt} = k(T - 22)$ . 1
- (ii) A wealthy industrialist is found murdered in his home. When police arrived on the scene at 11:00 pm, the temperature of the body was  $31^\circ\text{C}$ , and one hour later it was  $30^\circ\text{C}$ . The temperature of the room where the body was found was  $22^\circ\text{C}$ . Using Newton's law of cooling  $\frac{dT}{dt} = k(T - 22)$ , and the fact that normal body temperature is  $37^\circ\text{C}$ , estimate the time that the murder occurred. 3

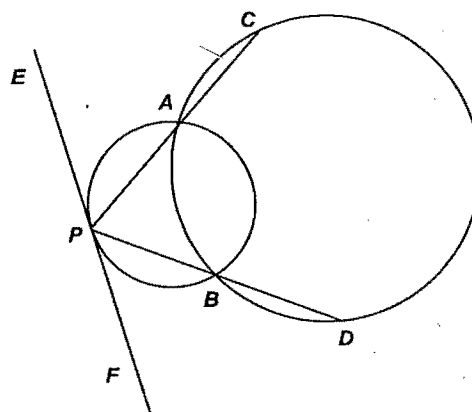
**[Question 4 continued]**

- (d) Copy the diagram into  
your answer booklet.

$PAC, PBD$  are straight lines.

$EF$  is the tangent at  $P$ .

Prove  $CD \parallel EF$



3

**Question 5 (12 marks)**

- (a) Use the substitution  $u = x - 1$  to evaluate  $\int_2^4 \frac{x}{(x-1)^2} dx$  3

- (b) A particle moves along a straight line in such a way that its distance  $x$  cm from a fixed point  $O$  at time  $t$  seconds is  $x = 2 \cos 3t$ .

- (i) Show that the particle is moving in Simple Harmonic Motion. 2
- (ii) Write down the period of the motion. 1
- (iii) Find the particle's speed when it is first 1 cm from  $O$ . 2

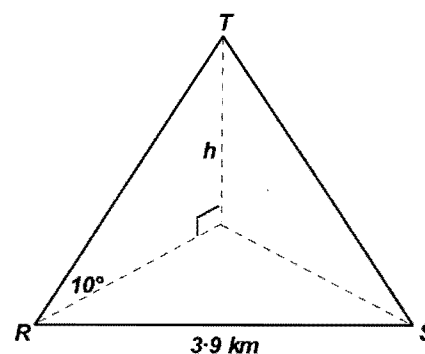
- (c) Two boats, *Rascal* and *Sirocco*, are sailing near Ball's Pyramid, a giant pillar of rock that rises from the sea. 4

The boats are 3.9 km apart, and the angle of

elevation of the top  $T$  of

Ball's Pyramid from *Rascal* is  $10^\circ$ .

Given that  $\angle TRS$  and  $\angle TSR$  are  $65^\circ$  and  $48^\circ$  respectively, find the height  $h$  of Ball's Pyramid to the nearest metre.



**Question 6 (12 marks)**

(a) Water is being pumped into an empty inverted conical tank – that is, a tank with its apex down – at a rate of  $300\,000\text{ cm}^3/\text{min}$ . The tank is 6 metres high and the diameter at the top is 4 metres. Let  $h$  be the height of the water and  $r$  the radius of the water surface at time  $t$  minutes.

Calculate the rate at which the water level is rising at the instant the water level reaches 2 metres. (Write your answer to 1 decimal place). 4

(b)  $P(2ap, ap^2)$  is a point on the parabola  $x^2 = 4ay$ , which has focus  $S$ .

(i) Derive the equation of the tangent to the parabola at  $P$ . Hence show that the tangent meets the x-axis at the point  $T(ap, 0)$ . 3

(ii) Find the coordinates of  $M$ , the point that divides  $ST$  externally in the ratio  $2 : 1$  2

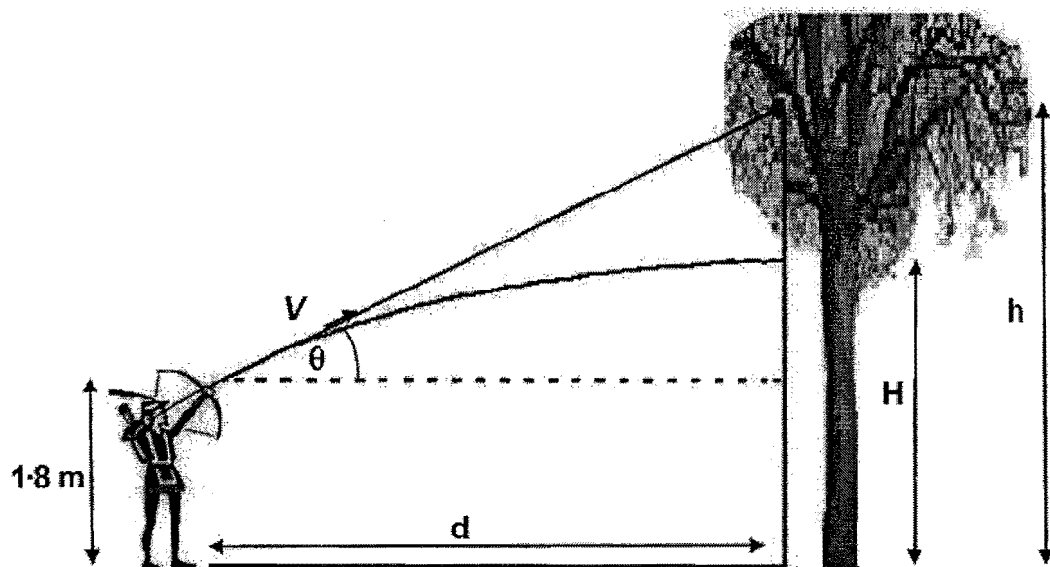
(iii) Describe geometrically the locus of  $M$  as  $P$  moves on the parabola. 1

(c) A function is given by the rule  $f(x) = \log_e \frac{1+x}{1-x}$ . Find the rule for the inverse function,  $f^{-1}(x)$ . 2

**Question 7 (12 marks)**

(a) Robin Hood aims his arrow at an acorn which is on an oak tree  $d$  metres away. The acorn is  $h$  metres above the ground, and Robin releases the arrow from a point 1.8 metres above the ground, at an angle of elevation of  $\theta$  degrees. Robin can vary the initial velocity  $V$  of the arrow.

At the instant Robin releases the arrow, the acorn begins to fall vertically downwards under gravity, with acceleration  $g$ . (Neglect air resistance).



**[Question 7 continued]**

- (i) Taking the ground at Robin's feet as the origin of coordinates, show that at time  $t$  the  $x$ - and  $y$ -coordinates of the arrow's tip are

$$x = Vt \cos \theta \quad \text{and} \quad y = Vt \sin \theta - \frac{gt^2}{2} + 1.8 \quad 3$$

- (ii) Show that, when the arrow reaches the tree, its vertical height above the ground is given by

$$H = d \tan \theta - \frac{gd^2 \sec^2 \theta}{2V^2} + 1.8 \quad 2$$

- (iii) Robin's arrow hits the acorn as it falls. Show mathematically why this will in fact always be the case in this situation, no matter what the initial velocity of the arrow.

3

(Provided, of course, that it is great enough to reach the tree!)

- (b) On a toss of two dice, John throws a total of 5. Find the probability that he will throw another 5 before he throws 7.

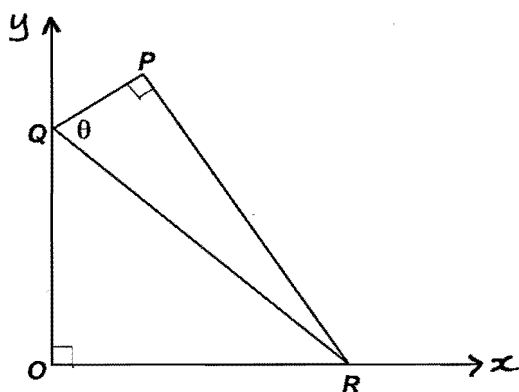
2

- (c) A triangle  $PQR$ , right angled at  $P$  and with  $\angle PQR = \theta$ , slides on a horizontal floor with its vertices  $Q$  and  $R$  in contact with perpendicular walls.

- (i) State why  $OQPR$  is a cyclic quadrilateral.
- (ii) Derive the equation of the locus of  $P$ .

1

1



QUESTION 1

(a)  $\lim_{x \rightarrow 0} \frac{\sin 3x}{x} = 3 \cdot \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$   
 $= 3 \cdot 1$   
 $= 3$

(b)  $\frac{d}{dx} \ln(\sin^3 x) = \frac{3 \sin^2 x \cdot \cos x}{\sin^3 x}$   
 $= \frac{3 \cos x}{\sin x}$   
 $= 3 \cot x$

(c)  $f(x) = x^2 - 6x + 10$   
 $= (x-3)^2 + 1$

The graph of  $y = f(x)$  is a concave up parabola, vertex (3,1).  $\therefore R: y \geq 1$

Alternatively,

$f'(x) = 2x - 6$   
 $= 0$  when  $x = 3$ .

Since  $f''(x) = 2 > 0$   $\therefore$  minimum turning point at  $x = 3, y = 1$ .  $\therefore R: y \geq 1$

(d)  $m_1 = \frac{1}{2}, m_2 = -3$

$\therefore \tan \theta = \left| \frac{\frac{1}{2} - (-3)}{1 + \frac{1}{2}(-3)} \right|$

$= 7$

$\therefore \theta = 81^\circ 52'$  (nearest minute)

Q1 cont'd

(e)  $\int_1^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx = \left[ \sin^{-1} \frac{x}{2} \right]_1^{\sqrt{3}}$   
 $= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2}$   
 $= \frac{\pi}{3} - \frac{\pi}{6}$   
 $= \frac{\pi}{6}$

(f) (i) Let  $x^3 - 4x^2 + 7x - 6 = P(x)$

Then  $P(2) = 8 - 16 + 14 - 6 = 0$

$\therefore (x-2)$  is a factor of  $P(x)$

(ii) 
$$\begin{array}{r} x^2 - 2x + 3 \\ x-2 \overline{) x^3 - 4x^2 + 7x - 6} \\ \underline{x^3 - 2x^2} \phantom{+ 7x - 6} \\ -2x^2 + 7x \phantom{- 6} \\ \underline{-2x^2 + 4x} \phantom{- 6} \\ 3x - 6 \\ \underline{3x - 6} \\ 0 \end{array}$$

$\therefore P(x) = (x-2)(x^2 - 2x + 3)$

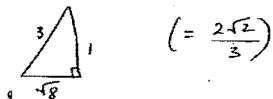
But  $x^2 - 2x + 3$  has  $\Delta = (-2)^2 - 4(1)(3) = -8 < 0$

and thus has no real roots.

$\therefore x = 2$  is the only real root.

QUESTION 2

(a)  $\cos \left[ \sin^{-1} \left( \frac{1}{3} \right) \right] = \frac{\sqrt{8}}{3}$



(b)  $3x^3 - 17x^2 - 8x + 12 = 0$

Let roots be  $\alpha, \beta, \gamma$ , and

Suppose  $\alpha\beta = 4$

Now  $\alpha\beta\gamma = -\frac{12}{3} = -4 \therefore \gamma = -1$

Also  $\alpha + \beta + \gamma = \frac{17}{3}$

$\therefore \alpha + \frac{4}{\alpha} - 1 = \frac{17}{3}$  (since  $\alpha\beta = 4$ )

$\therefore 3\alpha^2 - 20\alpha + 12 = 0$

$(3\alpha - 2)(\alpha - 6) = 0$

$\therefore \alpha = \frac{2}{3}$  or  $6$

ie. roots are  $-1, \frac{2}{3}, 6$ .

(c) Let  $\sqrt{12} \sin x + 2 \cos x = R \cos(x - \alpha)$

(i)  $= R \cos x \cos \alpha + R \sin x \sin \alpha$

$\therefore \left. \begin{array}{l} R \cos \alpha = 2 \\ R \sin \alpha = \sqrt{12} \end{array} \right\} \therefore R = 4$   
 $\tan \alpha = \frac{\sqrt{12}}{2} = \sqrt{3} \therefore \alpha = \frac{\pi}{3}$

$\therefore$  expression  $= 4 \cos(x - \frac{\pi}{3})$

(ii)  $4 \cos(x - \frac{\pi}{3}) = -3, 0 \leq x \leq 2\pi$

$\therefore \cos(x - \frac{\pi}{3}) = -\frac{3}{4}$

2.864

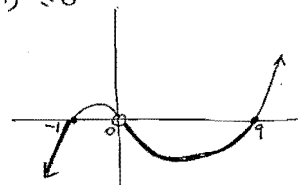
Q2 cont'd

(d)  $\frac{x^2 - 9}{x} \leq 8$

$\therefore x(x^2 - 9) \leq 8x^2 \quad (x \neq 0)$

$\therefore x[x^2 - 9 - 8x] \leq 0$

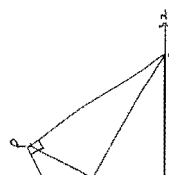
$x(x-9)(x+1) \leq 0$



$\therefore x \leq -1$  or  $0 < x \leq 9$ .

Q7 CONTINUED

(i) OQPR is a cyclic quadrilateral since opposite angles  $\angle OPR$  and  $\angle ORQ$  are supplementary (each  $= 90^\circ$ ).



Join PO. From O, PR is a chord of a circle. So  $\angle POQ = \angle POR$  (Angles in the same segment).

But this is true for all positions of P as Q, R slide along the walls.

$\therefore$  the gradient of PO is always  $\tan \theta$ .

Since PO goes through the origin, P's locus is the straight line  $y = (\tan \theta)x$ .

(ii)  $\therefore$  correct answer.

### QUESTION 3.

(a) (i) Let  $f(x) = \ln(x+1) + x - 1$

Then  $f(0) = \ln 1 + 0 - 1 = -1 < 0$

and  $f(1) = \ln 2 + 1 - 1 = \ln 2 > 0$

$f_0(0)$  and  $f(1)$  have opposite signs,  
hence a root lies between 0 and 1.

(ii)  $f'(x) = \frac{1}{x+1} + 1$

Let  $x_1 = \frac{1}{2}$

Then  $x_2 = \frac{1}{2} - \frac{f(\frac{1}{2})}{f'(\frac{1}{2})}$   
 $= \frac{1}{2} - \frac{(\ln 1.5 - 0.5)}{(\frac{1}{1.5} + 1)}$

$\approx 0.5567209$

$\approx 0.56$  (to 2 d.p.).

(b) (i)  $\frac{d}{dx} \left( \frac{1}{2} v^2 \right) = 32x(x^2+1)$   
 $= 32x^3 + 32x$

$\therefore \frac{1}{2} v^2 = 8x^4 + 16x^2 + C$

But  $x=0, v=4$

$\therefore \frac{1}{2} \cdot 4^2 = 0 + C \therefore C = 8$

$\therefore v^2 = 16x^4 + 32x^2 + 16$

$= 16(x^4 + 2x^2 + 1)$

$= 16(x^2+1)^2$

$\therefore v = \pm 4(x^2+1)$

### Q3 cont'd

(b) (ii) At  $t=0, x=0$  and  $v=4$ .

So particle is moving to the right. But since  
 $a > 0$  when  $x > 0$ , it will keep moving to  
the right.  $\therefore v > 0$

Thus  $v = 4(x^2+1)$

(iii)  $v = \frac{dx}{dt} = 4(x^2+1)$

$\therefore \frac{dt}{dx} = \frac{1}{4} \cdot \frac{1}{x^2+1}$

$t = \frac{1}{4} \tan^{-1} x + C$

But at  $t=0, x=0 \therefore C=0$

$\therefore 4t = \tan^{-1} x$

$\therefore x = \tan 4t$

(c) We prove that, for  $n \geq 1$ ,

$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}$

When  $n=1$

LHS  $= \frac{1}{1 \times 5} = \frac{1}{5}$

RHS  $= \frac{1}{4(1)+1} = \frac{1}{5}$

$\therefore$  it is true  
for  $n=1$

Assume it is true for  $n=k$  ( $k \geq 1$ )

$\therefore \frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} = \frac{k}{4k+1}$  (\*)

Thus

$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} + \frac{1}{[4(k+1)-3][4(k+1)+1]}$   
 $= \frac{k}{4k+1} + \frac{1}{[4(k+1)-3][4(k+1)+1]}$  by (\*)

### 3 cont'd

$= \frac{k}{4k+1} + \frac{1}{(4k+1)(4k+5)}$

$= \frac{k(4k+5) + 1}{(4k+1)(4k+5)}$

$= \frac{4k^2 + 5k + 1}{(4k+1)(4k+5)}$

$= \frac{(4k+1)(k+1)}{(4k+1)(4k+5)}$

$= \frac{(k+1)}{4k+5}$

Thus if true for  $n=k$ , it is also true  
for  $n=k+1$ .

$\therefore$  Since it is true for  $n=1$ , it is true  
for  $n=1+1=2$ , and thus also for  $n=2+1=3$ ,

and so on for all integers  $n \geq 1$ .

### QUESTION 4

(a)  $\int \cos^2 3x \, dx = \int \left( \frac{1}{2} + \frac{1}{2} \cos 6x \right) dx$   
 $= \frac{1}{2} x + \frac{1}{12} \sin 6x + C$

(b) (i) Graphing  $2x(1-x)$  versus  $x$  gives a

concave-down parabola

with axis of symmetry

$x = \frac{1}{2}$ .

$\therefore$  maximum value is  $2\left(\frac{1}{2}\right)\left(1-\frac{1}{2}\right) = \frac{1}{2}$

[Alternatively, Let  $f(x) = 2x(1-x)$

$\therefore f'(x) = 2 - 4x, f''(x) = -4$ .

For max.,  $f'(x) = 0 \therefore x = \frac{1}{2}$

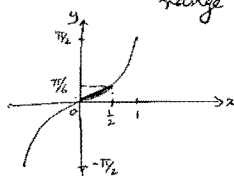
$f(x) = \frac{1}{2}$

Since  $f''(x) < 0 \therefore \frac{1}{2}$  is a maximum value.]

(ii) If  $0 \leq x \leq 1$  then  $0 \leq 2x(1-x) \leq \frac{1}{2}$  [from (i)]

Thus  $\sin^{-1}\{2x(1-x)\}$  has

range  $R: 0 \leq y \leq \frac{\pi}{6}$ .



✓ refers to  
 $a > 0$  since  
 $x > 0$

✓ correct sub.  
into correct  
formula

✓ correct  
answer

✓

✓ authentic  
simplification

✓

✓

✓ proves for  
 $n=1$

✓ correct  
start

✓ correct  
use of trig  
relation

✓ correct  
answer

✓ a correct  
demonstration

✓ relevant  
working

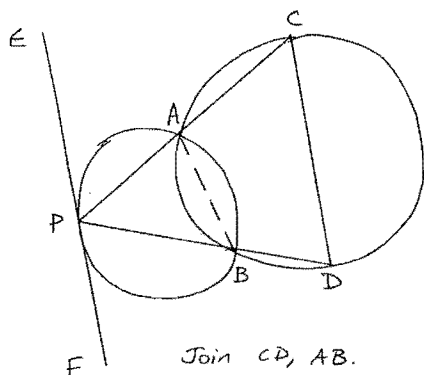
✓ correct  
answer

✓ correct  
working/  
simplifying

✓

Q4 cont'd

(d)



$\angle EPA = \angle ABP$  (alternate segment theorem) ✓

$\angle ABP = \angle ACD$  (exterior angle of cyclic quad. ACDB) ✓

$\therefore EF \parallel CD$  (alternate angles  $\angle EPA$  and  $\angle ACD$  are equal) ✓

Q4 cont'd

(c) (i) Let  $T = 22 + Ae^{kt}$

Then  $\frac{dT}{dt} = k \cdot Ae^{kt}$   
 $= k(T - 22)$

So  $T = 22 + Ae^{kt}$  is a solution.

(ii) Measure time  $t$  (in hours) from 11:00pm.

Thus at  $t=0$ ,  $T=31$

$t=1$ ,  $T=30$ .

That is,  $31 = 22 + Ae^0 \therefore A = 9$

and  $\therefore 30 = 22 + 9e^k$

$\therefore \frac{8}{9} = e^k$

$\therefore k = \ln\left(\frac{8}{9}\right) \left(\approx -0.117783\right)$  ✓ correct value

$\therefore 37 = 22 + 9e^{\ln(\frac{8}{9})t}$

$\therefore \frac{15}{9} = e^{\ln(\frac{8}{9})t}$

$\therefore t = \frac{\ln(\frac{15}{9})}{\ln(\frac{8}{9})}$

$= -4.337$

$\therefore$  he was murdered at  $(11 - 4.337)$  pm.

$= 6:40$  pm (nearest minute). ✓ correct answer

QUESTION 5

(a)  $\int_2^4 \frac{x}{(x-1)^2} dx$ . Let  $u = x-1$

$\therefore du = dx$   
 $x = u+1$

$x=2 \rightarrow u=1$   
 $x=4 \rightarrow u=3$

$\therefore \int = \int_1^3 \frac{u+1}{u^2} du$

$= \int_1^3 \left(\frac{1}{u} + u^{-2}\right) du$

$= \left[\ln u - \frac{1}{u}\right]_1^3$

$= \left(\ln 3 - \frac{1}{3}\right) - (\ln 1 - 1)$

$= \ln 3 + \frac{2}{3}$

$(\approx 1.765)$ . ✓ correctly set-up for u

(b) (i)  $x = 2 \cos 3t$

$\therefore \dot{x} = -6 \sin 3t$

$\therefore \ddot{x} = -18 \cos 3t$

$\therefore \ddot{x} = -9x$

This is in the form  $\ddot{x} = -n^2x$ , hence it's S.H.M.

(ii)  $T = \frac{2\pi}{3}$  seconds

(iii) When  $x=1$ ,  $1 = 2 \cos 3t$

$\therefore 3t = \frac{\pi}{3}$

$\therefore t = \frac{\pi}{9}$  seconds

At  $t = \frac{\pi}{9}$ ,  $|\dot{x}| = |-6 \sin 3(\frac{\pi}{9})| = 3\sqrt{3}$  cm/s. ✓ correct time

Q5 cont'd

(b) (iii) Alternatively,  $v^2 = n^2(a^2 - x^2)$

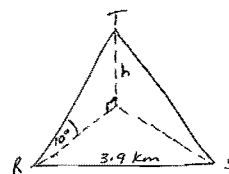
$= 9(4 - 1^2)$

$= 27$

$\therefore v = \pm\sqrt{27}$

$\therefore \text{speed} = |v| = 3\sqrt{3}$  cm/s ✓ correct use of formula

(c)



$\angle RTS = 180^\circ - (65^\circ + 48^\circ)$   
 $= 67^\circ$

$\therefore \frac{TR}{\sin 48^\circ} = \frac{3.9}{\sin 67^\circ}$

$\therefore TR = \frac{3.9 \sin 48^\circ}{\sin 67^\circ}$

But  $\frac{h}{TR} = \sin 10^\circ$

$\therefore h = \frac{3.9 \sin 48^\circ}{\sin 67^\circ} \cdot \sin 10^\circ$

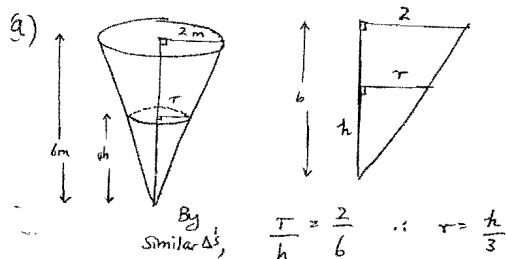
$\approx 0.5467$  km

$= 547$  m. (nearest metre). ✓ correct expression for TR

✓ correct expression for h

✓ correct answer

# QUESTIONS 6



$$\text{Similar } \Delta's, \quad \frac{r}{h} = \frac{2}{6} \quad \therefore r = \frac{h}{3}$$

$$\text{Thus } V = \frac{1}{3} \pi r^2 h \\ = \frac{1}{3} \pi \left(\frac{h}{3}\right)^2 h \\ \therefore V = \frac{\pi h^3}{27}$$

$$\text{Now } \frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}, \quad \left(\text{where } \frac{dV}{dh} = \frac{\pi h^2}{9}\right)$$

$$\therefore 300000 \frac{\text{cm}^3}{\text{min}} = \frac{\pi}{9} (200)^2 \text{ cm}^2 \cdot \frac{dh}{dt}$$

$$\therefore \frac{dh}{dt} = \frac{9}{\pi (200)^2} \cdot 300000 \frac{\text{cm}}{\text{min}} \\ \approx 21.4859 \\ = 21.5 \text{ cm/min. (1 d.p.)}$$

✓ expresses  $r$  in terms of  $h$

✓ correct expression for  $V$

✓ correct use of Chain Rule

✓ correct answer

## Q6 cont'd

$$(b) (i) \quad y = \frac{x^2}{4a} \quad \therefore \frac{dy}{dx} = \frac{x}{2a}$$

$$\therefore \text{at } P, \quad \frac{dy}{dx} = \frac{2ap}{2a} = p$$

$$\therefore \text{tangent is } y - ap^2 = p(x - 2ap)$$

$$\therefore y = px - ap^2$$

$$\text{At } T, y = 0 \quad \therefore px - ap^2 = 0 \\ x = \frac{ap^2}{p}$$

$$\therefore x = ap$$

$$\text{Hence } T = (ap, 0)$$

$$(ii) \quad S(0, a), T(ap, 0), \text{ let } k:l = 2:-1$$

$$\text{Then } M = \left[ \frac{(2)(ap) + (-1)(0)}{2 + (-1)}, \frac{2(0) + (-1)(a)}{2 + (-1)} \right]$$

$$\therefore M = (2ap, -a)$$

(iii) As  $P$  moves on the parabola,  $M$  moves along the line  $y = -a$ ;  $M$  moves along the directrix of the parabola.

$$(c) \text{ Let } y = f(x), \text{ i.e. } y = \log_e \left( \frac{1+x}{1-x} \right)$$

$$\text{For inverse, } x = \log_e \left( \frac{1+y}{1-y} \right)$$

$$\therefore e^x = \frac{1+y}{1-y}$$

$$e^x - e^x y = 1 + y$$

$$\therefore e^x - 1 = y(e^x + 1)$$

$$\text{Hence } y = \frac{e^x - 1}{e^x + 1}$$

✓ derives  $\frac{dy}{dx}$

✓ derives equation of tangent

✓ shows  $T = (ap, 0)$

✓ correct use of division formula

✓ correct coordinates of  $M$

✓ geometric answer

✓ correct to here

✓ correct answer

# QUESTION 7

$$\begin{aligned} (i) \quad \ddot{x} &= 0 & \ddot{y} &= -g \\ \dot{x} &= c & \dot{y} &= -gt + c \\ \text{at } t=0, \dot{x} &= V \cos \theta & \therefore c &= V \sin \theta \\ \therefore \dot{x} &= V \cos \theta & \therefore \dot{y} &= -gt + V \sin \theta \\ \therefore x &= Vt \cos \theta + c & \therefore y &= -\frac{gt^2}{2} + Vt \sin \theta + c \\ \text{at } t=0, x &= 0 \quad \therefore c = 0 & \text{at } t=0, y &= 1.8 \quad \therefore c = 1.8 \\ \therefore x &= Vt \cos \theta & \therefore y &= -\frac{gt^2}{2} + Vt \sin \theta + 1.8 \end{aligned}$$

Correct derivation + use of initial conditions

✓✓

$$(ii) \text{ "Arrow reaches tree" means } x = d \\ \therefore d = Vt \cos \theta \quad \therefore t = \frac{d}{V \cos \theta}$$

$$\therefore \text{at this time, } y = -\frac{g}{2} \left( \frac{d}{V \cos \theta} \right)^2 + V \left( \frac{d}{V \cos \theta} \right) \sin \theta + 1.8$$

$$\text{i.e. } H = -\frac{gd^2 \sec^2 \theta}{2V^2} + d \tan \theta + 1.8$$

✓ finds  $t$

✓ correct substitution + simplifying

✓ correct equations for acorn

$$\text{Thus at } t = \frac{d}{V \cos \theta}, \quad y = 1.8 - \frac{g}{2} \left( \frac{d}{V \cos \theta} \right)^2 \\ y = 1.8 - \frac{gd^2 \sec^2 \theta}{2V^2}$$

$$\text{But } \frac{h-1.8}{d} = \tan \theta \quad \therefore h = d \tan \theta + 1.8$$

$$\text{Hence } y = d \tan \theta + 1.8 - \frac{gd^2 \sec^2 \theta}{2V^2} = H$$

i.e. the acorn is the same height above the ground as the arrow arrives at the tree.

✓ writes  $h$  in terms of  $d$

✓ correct conclusion

## Q7 cont'd

(b) John will succeed if he throws a total of 5 on the next toss, or should he not throw a 5 or 7 on this toss but throws 5 on the next; or should he not throw a 5 or 7 on either of these tosses but throw 5 on the next; etc.

$$\begin{aligned} \therefore P(\text{throws 5 before 7}) &= \frac{4}{36} + \frac{26}{36} \cdot \frac{4}{36} + \left( \frac{26}{36} \right)^2 \cdot \frac{4}{36} + \dots \\ &= \frac{\frac{4}{36}}{1 - \frac{26}{36}} \quad \left( \text{since } S_{\infty} = \frac{a}{1-r} \right) \\ &= \frac{2}{5} \end{aligned}$$

✓ correct logic + series

✓ correct answer

[Alternatively,

John keeps throwing until he gets either a 5 or a 7. We need to find the probability the 5 comes up before the 7.

So the question is, given he gets a 5 or a 7, what is the probability that it is a 5?

There are 10 ways to throw a 5 or a 7, and of these four give a 5.

So the probability he gets a 5 before a 7 is just  $\frac{4}{10} = \frac{2}{5}$ .

✓✓

(no marks for just finding  $P(5 \text{ or } 7)$  on a single throw).

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